

P R O X M O X V E

Automated Remote SSH Shutdown

Technical Guide for Fully Automatic, Zero-Interaction Host Shutdown

Document Version	1.6
Date	October 2026
Aligned with	Proxmox VE 9.2 (official Administration Guide & qm/pct man pages)
Classification	Internal Use

1. Introduction

This document provides a complete, production-ready procedure for performing a fully automated remote shutdown of a Proxmox VE host via SSH. The procedures and commands are aligned with the official Proxmox VE 9.2 Administration Guide and the qm / pct manual pages.

Two practical methods are described:

- Remote-executed method (recommended for explicit control)** — The remote system issues all guest shutdown commands over SSH and then powers off the host. No additional script is stored on the Proxmox node.
- Simple method (official built-in behaviour)** — The remote system triggers a standard host shutdown. Proxmox automatically stops guests according to the configured Start/Shutdown Order via the pve-guests service.

2. Official Proxmox VE Shutdown Behaviour

According to the official Proxmox VE documentation and implementation:

- Issuing shutdown, poweroff, or the GUI Shutdown button on a node triggers the systemd unit pve-guests.service.
- This service executes pvesh create /nodes/localhost/stopall, which stops all running guests.
- Guests are stopped in the reverse of their configured Start/Shutdown Order.
- For QEMU/KVM virtual machines, Proxmox first attempts a shutdown via the QEMU Guest Agent (if enabled and running). If that is not available, an ACPI power-down signal is sent. After the configured timeout the guest is force-stopped.
- For LXC containers a SIGPWR signal is sent to the container's init process.

3. Guest Start / Shutdown Order

The Start/Shutdown Order is defined per guest and is fully supported by the official qm and pct tools (see man qm and the Administration Guide).

3.1 Official Behaviour

- Start order is ascending** (1 → 2 → 3). Guests with lower numbers start first.
- Shutdown order is the reverse** of the start order (highest number first).
- Guests that share the same order value are processed approximately in parallel.
- Guests with no order defined start last and are shut down first.

3.2 Recommended Priority

1. **Database / Storage services** → Order = 1 (start first, shut down last)
2. **Application / Middleware services** → Order = 2
3. **Frontend / Web / Monitoring services** → Order = 3 (start last, shut down first)

3.3 Configuration (Official Commands)

Web interface: Select the VM or CT → Options → Start/Shutdown Order

Command-line (from official man pages):

```
qm set <vmid> --startup order=1,up=30,down=180
pct set <ctid> --startup order=2,up=10,down=60
```

4. QEMU Guest Agent (Official Recommendation)

The official Proxmox documentation strongly recommends enabling the QEMU Guest Agent for the most reliable graceful shutdown of virtual machines. When the Guest Agent is enabled and running inside the guest, Proxmox uses it instead of (or in preference to) a pure ACPI signal.

Required steps:

1. Install the qemu-guest-agent package inside the guest operating system.
2. Ensure the agent service is running and enabled at boot.
3. In the Proxmox web interface go to the VM → Options and set “QEMU Guest Agent” to Enabled. A full shutdown/start of the VM is required for the change to take effect.

5. SSH Authentication and Limited User

A dedicated, non-privileged user is created on the Proxmox host. Authentication is performed exclusively with an SSH key that has an empty passphrase, enabling fully unattended operation.

5.1 Proxmox Host Configuration

1. Create the limited user:

```
adduser --disabled-password --gecos "" shutdownuser
```

2. Install sudo (not included by default on Proxmox):

```
apt update && apt install sudo -y
```

3. Grant the minimum required privileges (edit with visudo):

```
Defaults:shutdownuser !requiretty
shutdownuser ALL=(root) NOPASSWD: /usr/sbin/qm, /usr/sbin/pct, /usr/sbin/shutdown,
/sbin/shutdown, /usr/sbin/poweroff, /sbin/poweroff
```

Important: The !requiretty option is mandatory for non-interactive SSH sessions.

5.2 Remote Host Configuration

4. Generate an SSH key pair with an empty passphrase:

```
ssh-keygen -t ed25519 -f ~/.ssh/proxmox_shutdown -N ""
```

5. Install the public key on the Proxmox host:

```
ssh-copy-id -i ~/.ssh/proxmox_shutdown.pub shutdownuser@PROXMUX_IP
```

6. Correct the permissions on the Proxmox host:

```
chmod 700 /home/shutdownuser/.ssh
chmod 600 /home/shutdownuser/.ssh/authorized_keys
chown -R shutdownuser:shutdownuser /home/shutdownuser/.ssh
```

6. Recommended Method — Remote-Executed Shutdown

In this approach the remote automation system issues every required command over SSH. No additional script file is placed on the Proxmox host. The commands used (qm shutdown, pct shutdown, qm stop, pct stop, shutdown) are the official tools documented in the Proxmox man pages.

Advantages

- No additional files or packages are required on the Proxmox node
- Logic remains under the control of the remote automation platform
- Full control of sequencing, timeouts and logging
- Native compatibility with Home Assistant shell_command, NUT, systemd timers and cron

6.1 Reference Script (Remote Side)

Place the following script on the remote automation host:

```
#!/bin/bash
# proxmox-remote-shutdown.sh
set -e
PROXMOX="shutdownuser@PROXMOX_IP"
KEY="-i $HOME/.ssh/proxmox_shutdown -o BatchMode=yes -o StrictHostKeyChecking=no"
SSH="ssh $KEY $PROXMOX"

echo "$(date) - Starting remote graceful shutdown"

# 1. Shut down all running VMs (official qm shutdown)
echo "Shutting down VMs..."
VMS=$(($SSH "sudo qm list | awk 'NR>1 && \$3==\"running\"' {print \$1}'")
for vmid in $VMS; do
    echo "    qm shutdown $vmid"
    $SSH "sudo qm shutdown $vmid --timeout 120" &
done
wait

# 2. Shut down all running containers (official pct shutdown)
echo "Shutting down containers..."
CTS=$(($SSH "sudo pct list | awk 'NR>1 && \$2==\"running\"' {print \$1}'")
for ctid in $CTS; do
    echo "    pct shutdown $ctid"
    $SSH "sudo pct shutdown $ctid --timeout 60" &
done
wait

# 3. Force-stop any remaining guests (official qm/pct stop)
echo "Waiting 30 seconds..."
sleep 30

echo "Force-stopping remaining VMs..."
VMS=$(($SSH "sudo qm list | awk 'NR>1 && \$3==\"running\"' {print \$1}'")
for vmid in $VMS; do
    echo "    qm stop $vmid"
    $SSH "sudo qm stop $vmid"
done

echo "Force-stopping remaining containers..."
CTS=$(($SSH "sudo pct list | awk 'NR>1 && \$2==\"running\"' {print \$1}'")
for ctid in $CTS; do
    echo "    pct stop $ctid"
    $SSH "sudo pct stop $ctid"
done

# 4. Power off the host (official shutdown)
echo "All guests handled. Shutting down host..."
$SSH "sudo /usr/sbin/shutdown -h now"
```

Make the script executable:

```
chmod +x proxmox-remote-shutdown.sh
```

7. Simple Method — Official Built-in Guest Order

This method relies entirely on the official Proxmox behaviour. When a host shutdown is issued, the pve-guests service automatically stops all running guests according to the configured Start/Shutdown Order and the down= timeouts. This is the same sequence used by the web-interface Shutdown button.

```
ssh -i ~/.ssh/proxmox_shutdown -o BatchMode=yes shutdownuser@PROXMOX_IP 'sudo /usr/sbin/shutdown -h now'
```

8. Security Considerations and Testing

- Disable password authentication in the SSH daemon configuration.
- Restrict SSH access to trusted source IP addresses using the Proxmox firewall.
- Perform an initial test using a delayed shutdown (for example: shutdown -h +2).
- On HA-enabled clusters, consider placing the node into maintenance mode or stopping the HA services before a planned shutdown.

9. Recommended Implementation Path

1. Enable the QEMU Guest Agent on all important virtual machines (Section 4).
2. Configure the Start/Shutdown Order for all critical guests (Section 3).
3. Create the limited user, install the SSH key and grant the required sudo privileges (Section 5).
4. **Deploy the remote-executed method** described in Section 6 (or the simple official method in Section 7).
5. Trigger the shutdown from Home Assistant, NUT, a monitoring system or a scheduled task using a single non-interactive SSH call.

This document is fully aligned with the official Proxmox VE 9.2 Administration Guide and man pages.